JAMES CHRISTOPHER WHITMORE

Oxford, United Kingdom

james.whitmore@gmail.com

+44 7891 452367

linkedin.com/in/jameswhitmore

github.com/jwhitmore

PROFESSIONAL SUMMARY

Senior Research Engineer with 6 years of experience developing machine learning systems and distributed computing solutions. Passionate about tackling complex algorithmic problems and mentoring junior engineers. Spent the last three years building scalable data pipelines at high-growth startups, with particular expertise in deep learning inference optimization and cloud infrastructure.

WORK EXPERIENCE

Senior Research Engineer

Quantum Dynamics Lab

quantumdynamics.io

Oxford, United Kingdom

June 2022 - Present

Architected and deployed a distributed training framework that reduced model training time by 40% for convolutional neural networks across GPU clusters. Led a team of two junior engineers in designing novel data augmentation strategies for computer vision tasks, resulting in 8% improvement in model accuracy. Conducted research and published findings on efficient transformer model compression, contributing to two peer-reviewed conference papers. Implemented automated benchmarking pipeline for evaluating emerging deep learning frameworks against existing production systems.

Research Engineer

Nexus Analytics Limited

nexus-analytics.co

London, United Kingdom

August 2019 - May 2022

Developed and maintained core machine learning infrastructure supporting real-time prediction services for 50+ clients, handling 2 million daily inference requests. Designed and implemented a custom gradient descent optimization algorithm that improved convergence speed by 35% for sparse datasets. Built comprehensive monitoring and alerting systems for production ML models using Prometheus and Grafana. Collaborated with data science team to transition three legacy Python scripts into scalable microservices deployed on Kubernetes.

Machine Learning Engineer

DataFlow Systems

dataflow-systems.net

Cambridge, United Kingdom

July 2018 - July 2019

Implemented neural network models for time series forecasting in energy consumption prediction, achieving RMSE improvements of 22% over baseline methods. Created documentation and code examples for internal ML toolkit used by 15 engineers across the organization. Optimized database queries and implemented caching strategies that reduced API response times from 800ms to 150ms. Conducted code reviews and participated in designing

architectural improvements for batch processing pipeline handling 500GB daily data ingestion.

EDUCATION

Doctor of Philosophy in Computer Science

University of Oxford

Oxford, United Kingdom

2015 - 2018

Bachelor of Science in Mathematics with Computer Science

University of Manchester

Manchester, United Kingdom

2011 - 2015

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, Go, SQL, Bash

Machine Learning: TensorFlow, PyTorch, scikit-learn, XGBoost, Keras

Data Processing: Apache Spark, Pandas, NumPy, Dask

Cloud Platforms: AWS, Google Cloud Platform, Docker, Kubernetes

Databases: PostgreSQL, MongoDB, Redis, Elasticsearch

Tools and Version Control: Git, Jupyter Notebooks, Linux, Airflow

PUBLIC REPOSITORIES

github.com/jwhitmore/transformer-compression - PyTorch implementation of model pruning and quantization techniques for efficient transformer deployment

github.com/jwhitmore/distributed-training-framework - Custom framework for distributed deep learning across multi-GPU clusters with automatic fault tolerance

github.com/jwhitmore/time-series-forecasting - Production-grade LSTM models for energy consumption prediction with evaluation metrics and deployment scripts

github.com/jwhitmore/neural-network-optimization - Research implementations of various gradient descent variants and adaptive learning rate algorithms